Anmol Rawat

anmolrawat1509@gmail.com | +91-7906805660 | linkedin.com/in/anmol-rawat-2a691a288 | github.com/ANMOLRAWAT990

EDUCATION

Bachelor of Technology: Computer Science & Engineering **2023 – 2027**
Graphic Era Deemed to be University CGPA: **9.20 / 10.0**

Class XII: PCM **2023**
Beverly Hills Shalini School 80.60%

Class X **2021**
Kendriya Vidyalaya IIP 96.50%

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, C, Java
Backend: FastAPI, Flask, REST APIs, JWT, Role-Based Access Control (RBAC)
Frontend: React (Vite), Tailwind CSS, HTML5/CSS3
Databases: MongoDB, MySQL, SQLite
ML / AI / CV: scikit-learn, XGBoost, GaussianHMM, TensorFlow/Keras (LSTM), YOLOv8, OpenCV, Google Gemini API
Data Engineering: Parquet, pandas, NumPy, CCXT (Market Streams), Optuna (Hyperparameter Tuning)
Systems: Multithreading, ThreadPoolExecutor, Applied Cryptography (AES, PBKDF2), Task Scheduling
DevOps & Tools: Docker, Git, GitHub, Linux/Bash, XAMPP
Concepts: Data Structures & Algorithms, OOP, System Design, Time-Series Modeling

EXPERIENCE

Software Engineering Intern: AI Systems: TBI-GEU **Jun 2026 – Aug 2026**
SentiNaut: B2B Reputation & Operations SaaS for Hospitality • React (Vite), Tailwind CSS, Python/FastAPI, MongoDB, Google Gemini API, Razorpay

- Spearheaded architecture and development of a full-stack B2B SaaS platform with **3 role-based dashboards** and a Gemini-powered review pipeline classifying batches of **20** reviews with automatic retry logic.
- Designed a multi-tiered RBAC schema in MongoDB with JWT middleware and rate-limiting, isolating data access by property and role.
- Integrated **Razorpay** for subscription and plan-gating across single and multi-property tiers, with automated portfolio-wide upgrade sync.
- Engineered an analytics layer for **7/30-day** rolling sentiment trends and SLA tracking, plus a Guest CRM with WhatsApp deep-linking; abstracted the AI provider behind a `BaseAIProvider` interface to decouple business logic from the LLM vendor.

PROJECTS

Multi-Asset Quantitative Trading Engine (OmegaTrade-AI) **Jan 2026 – Present**
Python, XGBoost, TensorFlow/Keras, GaussianHMM, FastAPI, Docker, Optuna, CCXT, Parquet

- Developed a **4-agent ensemble** (XGBoost, GaussianHMM regime detector, CNN-LSTM, rule-based structure agent) with a meta-learner probabilistically weighting votes, fed by continuous CCXT polling of **50,000+ candles** cached in Parquet.
- Optimized model generalization via **Optuna** hyperparameter search with time-series cross-validation, replacing leakage-prone static thresholds with volatility-scaled ATR targets.
- Implemented a guarded auto-retraining safeguard validating candidate models against the live incumbent before atomic promotion, extending multi-asset coverage via a Dockerized FastAPI backend with Kelly-criterion sizing.

Road Hazard Detection: Real-Time CV Alert System **Mar 2026 – May 2026**
Python, Flask, YOLOv8 (Ultralytics), OpenCV, PHP, MySQL

- Curated a dataset of **400+** pothole and speed-bump images with custom Roboflow bounding-box annotations, augmenting with multi-angle variants to improve detection robustness.
- Fine-tuned a YOLOv8 detector deployed via a Flask inference server (XAMPP/PHP frontend) with confidence-gated (**>60%**) real-time alerting and bcrypt-hardened authentication.

High-Throughput Parallel Encryption Engine with ML Scheduler & Key Vault **Jul 2025 – Nov 2025**
Python, AES-CBC, SQLite, PBKDF2HMAC, scikit-learn, EWMA, ThreadPoolExecutor

- Constructed a SQLite-backed Key Vault deriving a master key via **PBKDF2HMAC-SHA256 (200,000 iterations)**, generating unique salts and IVs per file to mitigate replay attacks.
- Trained a scikit-learn cost model on file size and entropy to predict encryption time, with an EWMA fallback for unseen file types.
- Devised a heap-based scheduler with entropy profiling to distribute workloads across a `ThreadPoolExecutor`, cutting straggler idle time by **~30%**.